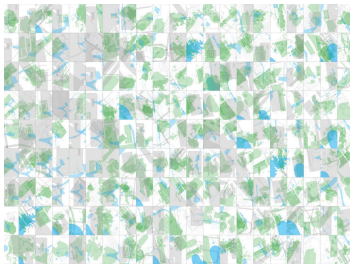
PORTFOLIO · CODE & AI

JOHN NORTHRUP

Designer & Digital Fabrication Specialist | Computational Workflows & Physical Systems



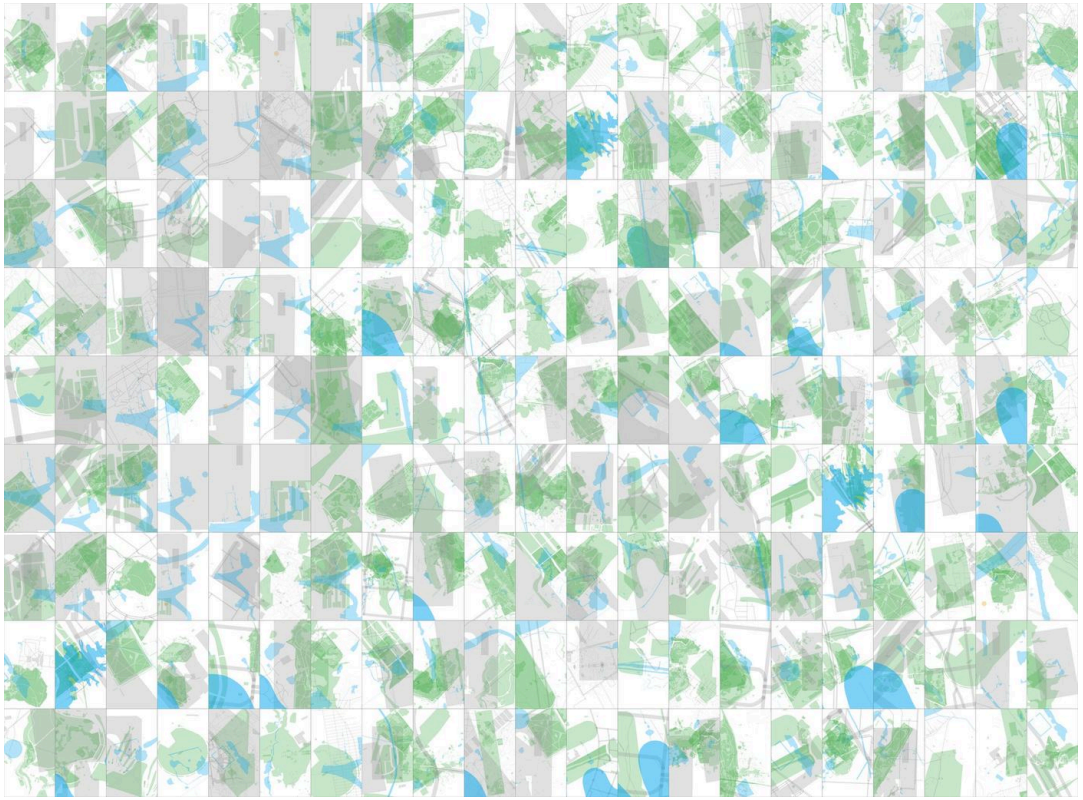


I am a designer and technical fabricator specializing in computational modeling, digital workflows, and physical construction. Grounded in hands-on material fabrication—from fine joinery to structural welding—and technical spatial design, my work bridges complex digital systems with buildable, physical space. I focus on technical design development, project management, and translating complex digital models into precise physical outcomes.

JCNORTHRUP@GMAIL.COM JCNORTHRUP.COM

CODE & AI

01	memoryMACHINE	SU26, M.ARCH THESIS	03
02	AMENITY DECK ENGINE	2026, SELF-DIRECTED COMPUTATIONAL STUDY	06
03	PROTOTYPE	SP26, PROTOTYPE, 3GB STUDIO	14



01

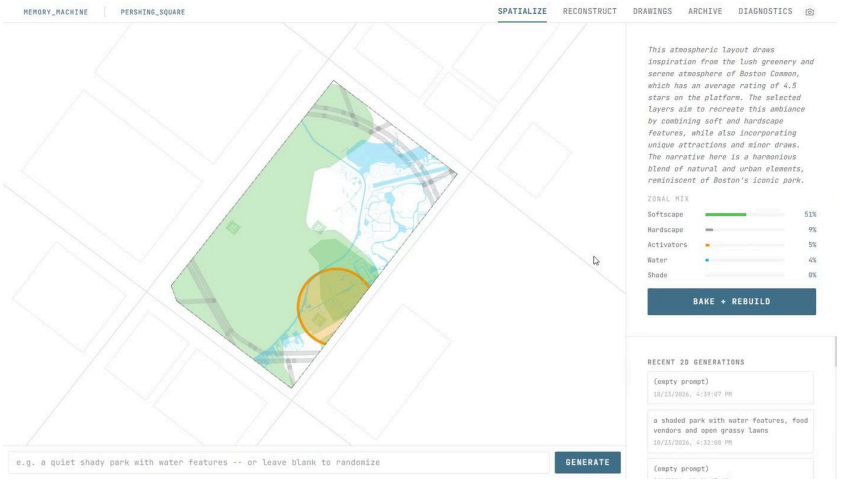
memoryMACHINE

SU26, M.ARCH THESIS
ADVISOR · CASEY REHM

memoryMACHINE is a computational design framework that transforms civic sentiment and archival memory into operative spatial data to revitalize under-utilized public space. Using Pershing Square in Downtown Los Angeles as a test bed—a civic site defined by a history of top-down, disconnected redesigns—the project proposes an adaptive spatial infrastructure in place of the fixed, one-time master plan.

At its core, a multimodal data ingestion engine aggregates web reviews, archival records, and spatial metrics, pairing qualitative sentiment with physical variables such as microclimate, canopy coverage, and circulation flows. To ensure equitable representation, the engine applies demographic weighting to mitigate digital access bias. These inputs are translated into spatial density fields and programmatic adjacencies, generating interventions—such as terraced landforms, water systems, and subterranean transit connections—as evidence-based spatial responses rather than predetermined formal gestures.

By merging expert analysis with experiential civic data, memoryMACHINE repositions the architect from sole author to system mediator. Public input shifts from a one-off consultation phase into a permanent, operative design parameter. The result is a self-regulating spatial framework that ingests usage patterns to continuously re-evaluate the site and adapt through phased physical interventions—establishing a democratic model in which public space evolves alongside the community it serves.



SOFTWARE | GENERATE A PLAN, REBUILD IT AS MASSING, DRAW IT (SPED UP) (VIDEO: JCNOR...



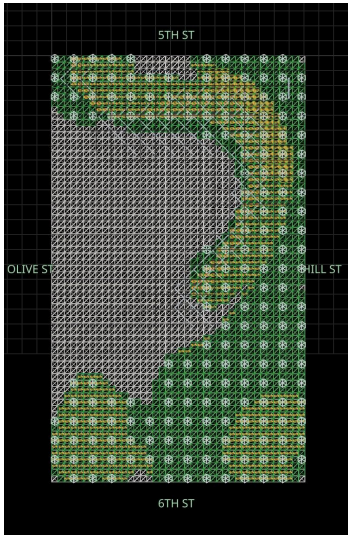
INSTALLATION | TIMBER SITE MODEL IN THE EXHIBITION HALL



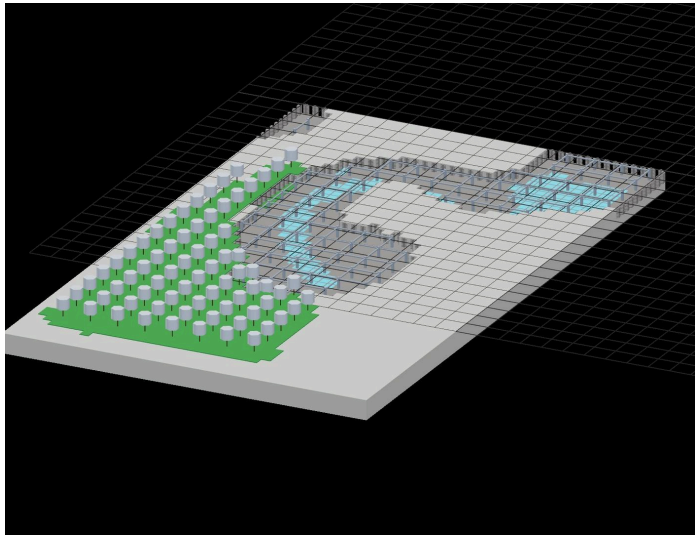
INSTALLATION | TIMBER SITE MODEL



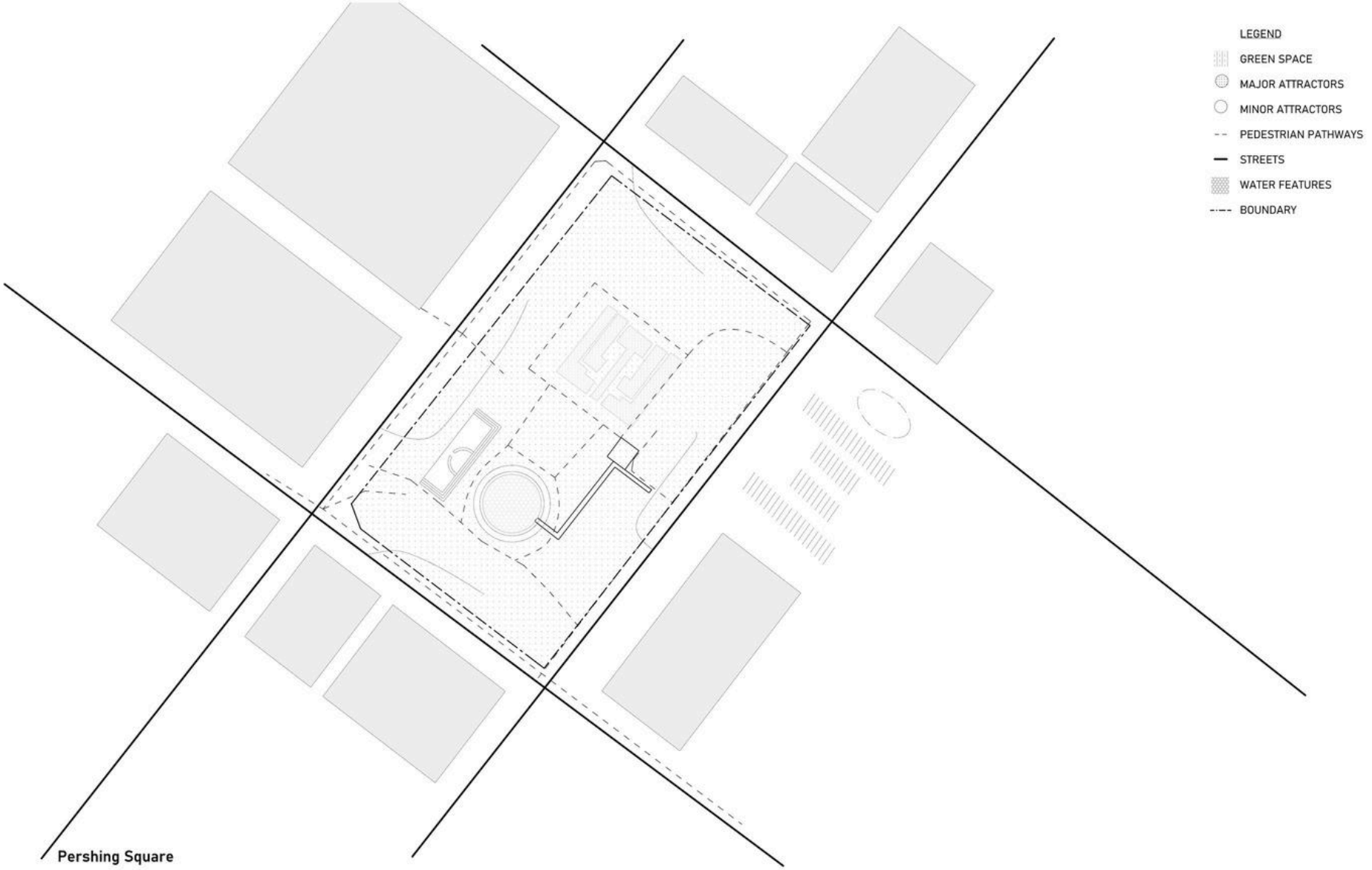
RENDER | THE GENERATED PARK IN ITS DOWNTOWN CONTEXT



GENERATED PLAN | PERSHING SQUARE



GENERATED MODEL | THE SAME SCHEME AS A MASSING



DRAWING | PERSHING SQUARE, LOS ANGELES: THE PROJECT SITE READ IN THE SYSTEM'S OWN TERMS



02

AMENITY DECK ENGINE

2026, SELF-DIRECTED COMPUTATIONAL STUDY
ROLE · DESIGN & SCRIPTING

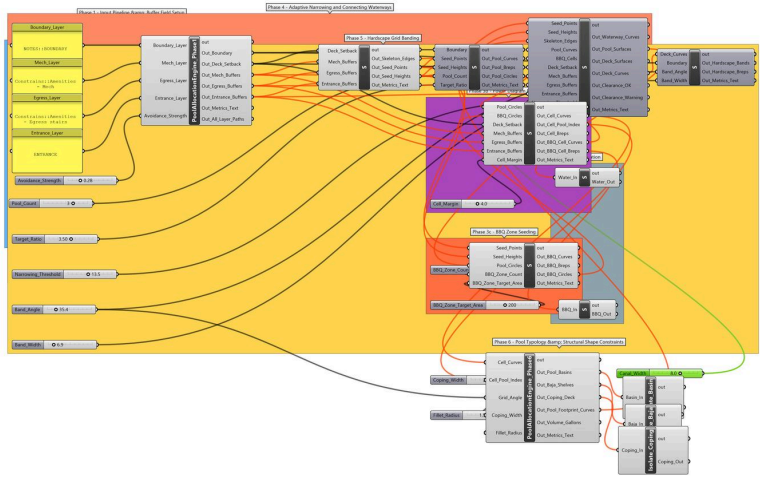
A parametric workflow for a high-rise podium deck, balancing complex site constraints—mechanical clearances, egress cores, and hardscape—with three distinct privacy zones.

The engine is a six-phase Grasshopper and Python pipeline. It reads the Rhino boundary, buffers the fixed obstacles, and uses a straight-skeleton algorithm to seed pool locations against a target deck-to-water ratio. A weighted Voronoi power diagram resolves the overlaps into obstacle-clipped cells, and a final phase converts those cells into grid-aligned basins with coping, typed by width.

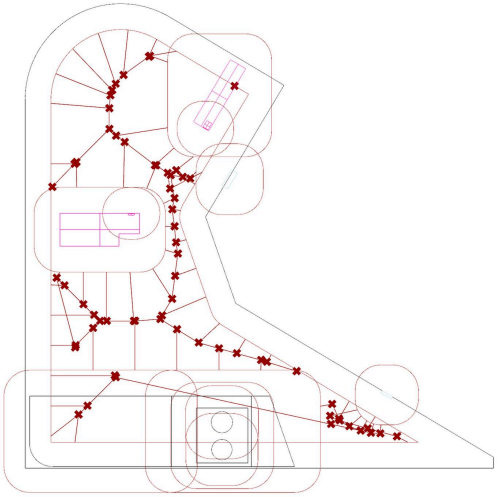
What it returns is not a finished plan but an allocation—curves and cells showing where programmatic density wants to settle on a tightly constrained floor plate. Re-run, one definition produces a field of options rather than a single answer.

The deck was designed off that allocation. I offset from the curves the engine produced and developed the basins, coping and soaking pools by hand: not arbitrarily, but from the shapes the script gave me. The engine did the spatial reasoning; the design judgment was mine. Every drawing and diagram here is output from the definition.

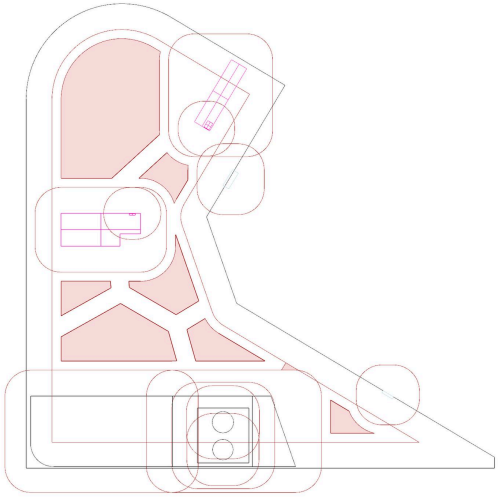
THE ENGINE



DEFINITION | SIX-PHASE PIPELINE, GRASSHOPPER AND PYTHON

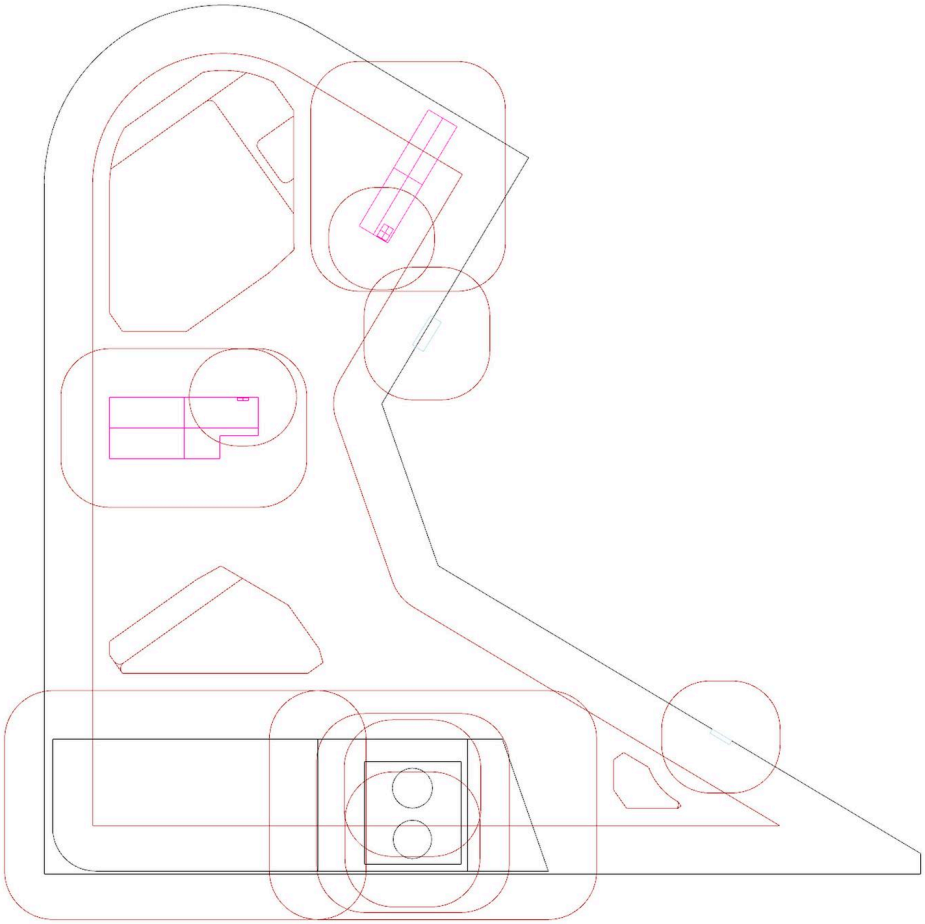


PHASE 2 | STRAIGHT SKELETON AND SEED POINTS

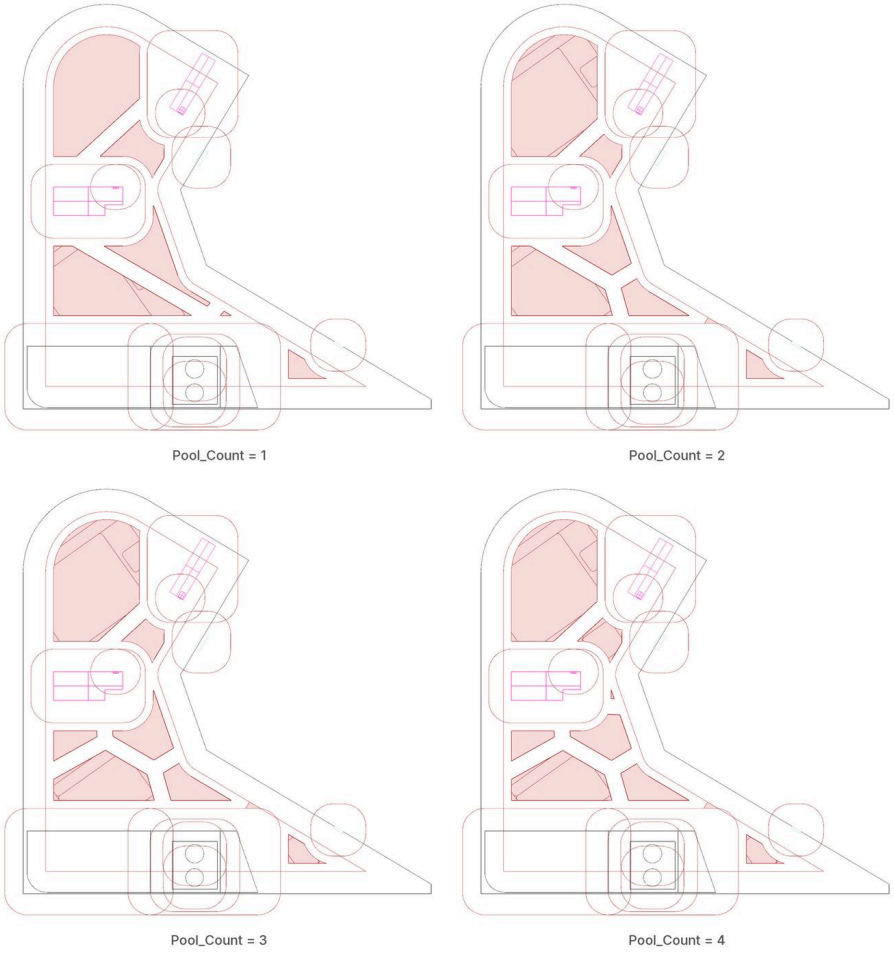


PHASE 3B | WEIGHTED VORONOI POWER DIAGRAM

THE ENGINE

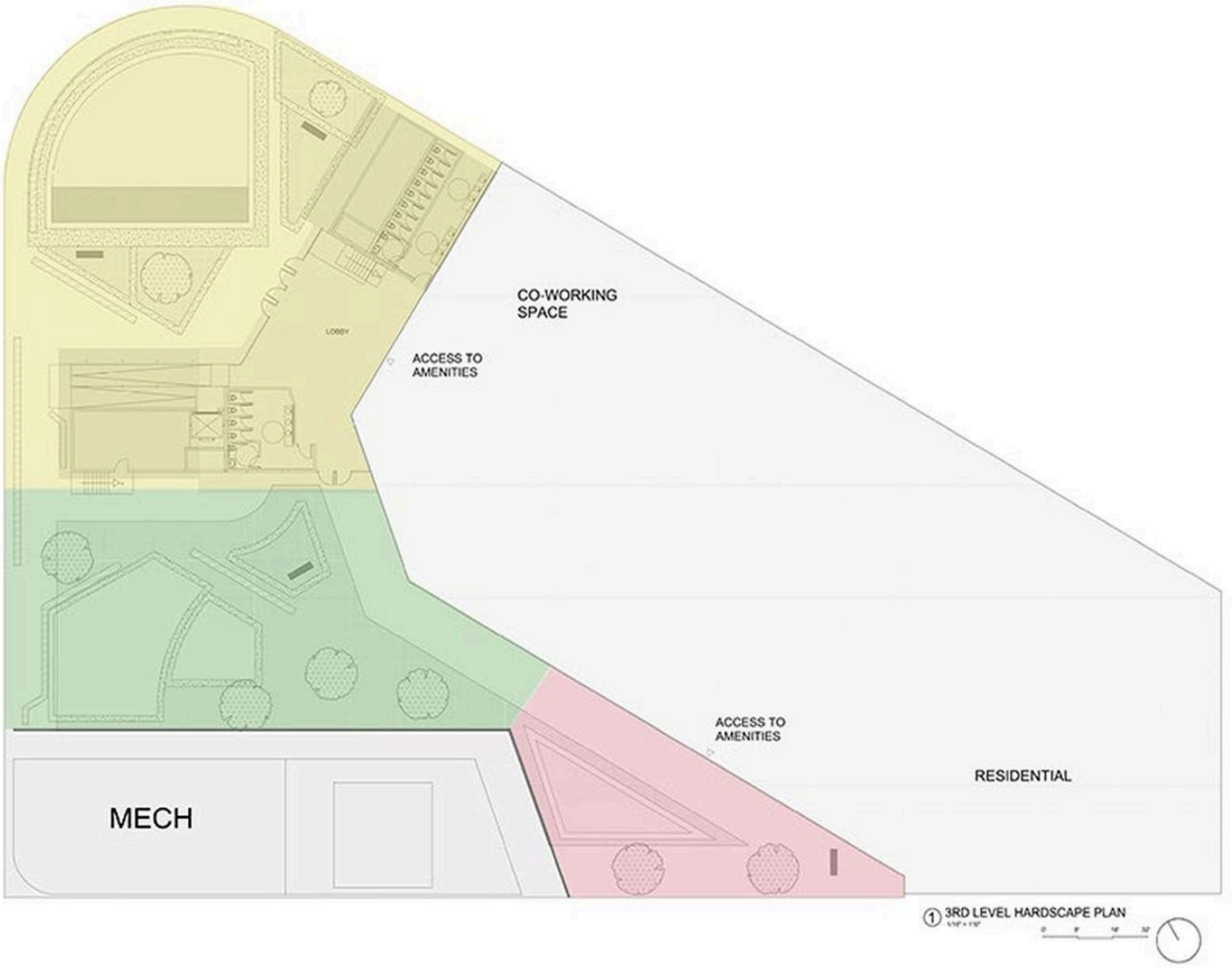


PHASE 6 | CELLS CONVERTED TO BUILDABLE BASINS



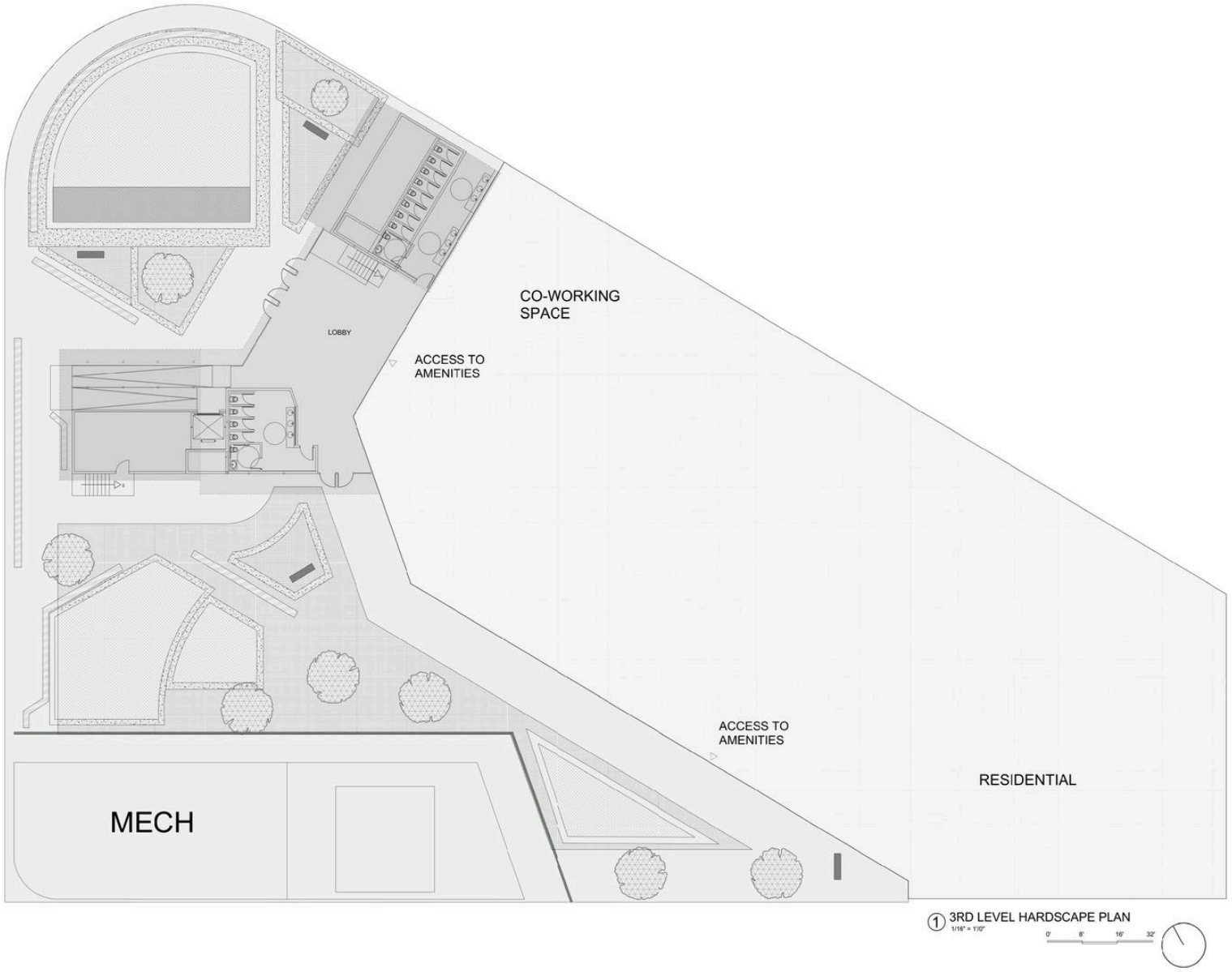
OPTION ARRAY | ONE DEFINITION, FOUR POOL COUNTS

FROM ALLOCATION TO DESIGN



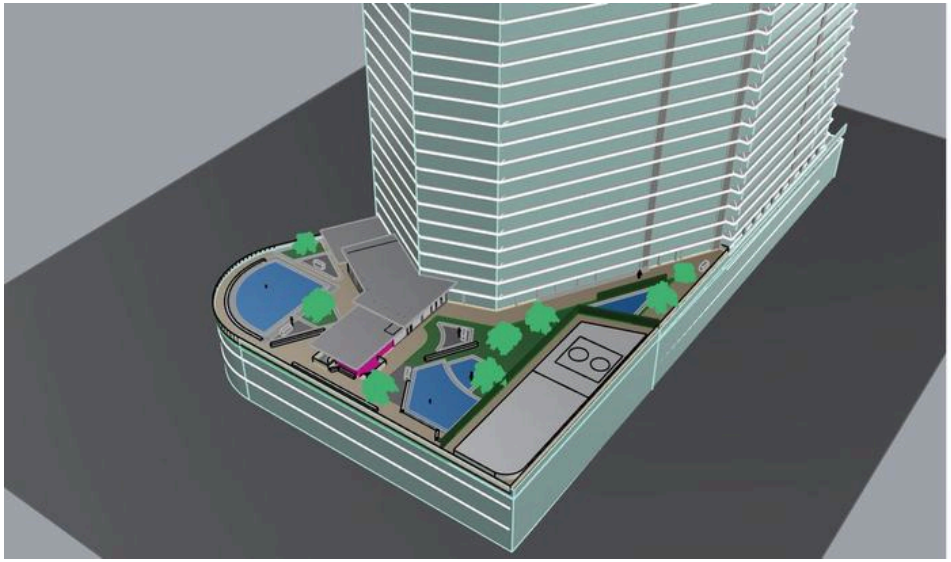
DESIGN NARRATIVE | THREE PRIVACY ZONES

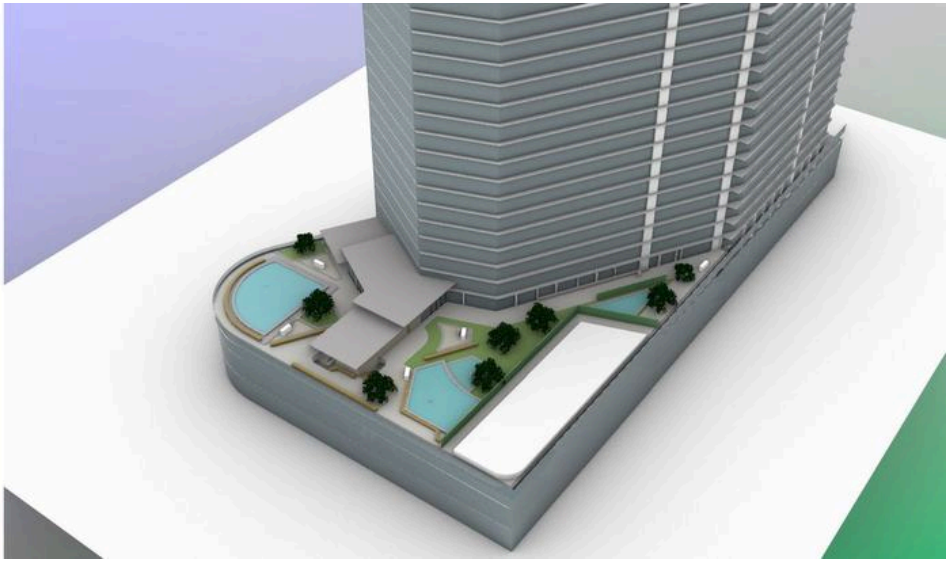
THE DECK



DRAWING | HARDSCAPE PLAN, THE DELIVERABLE

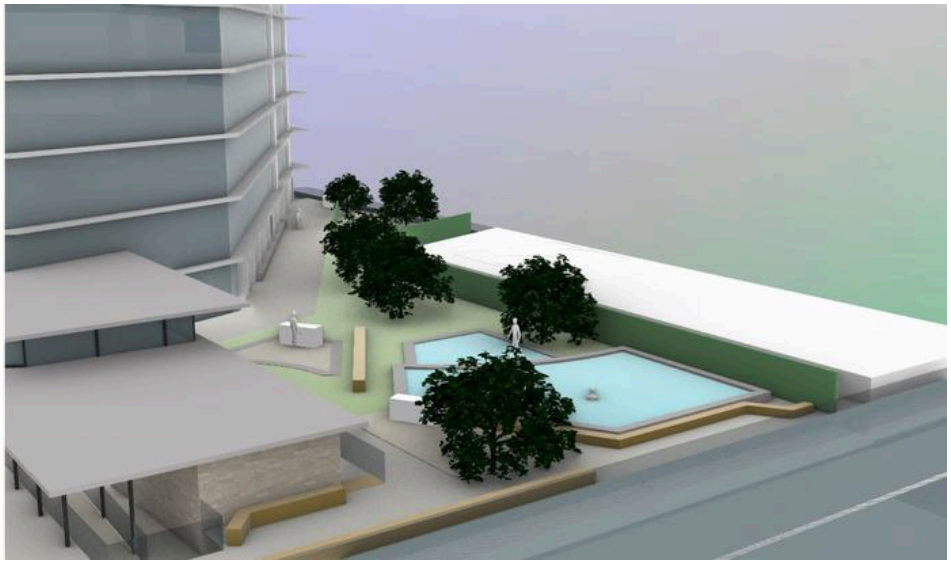
THE DECK





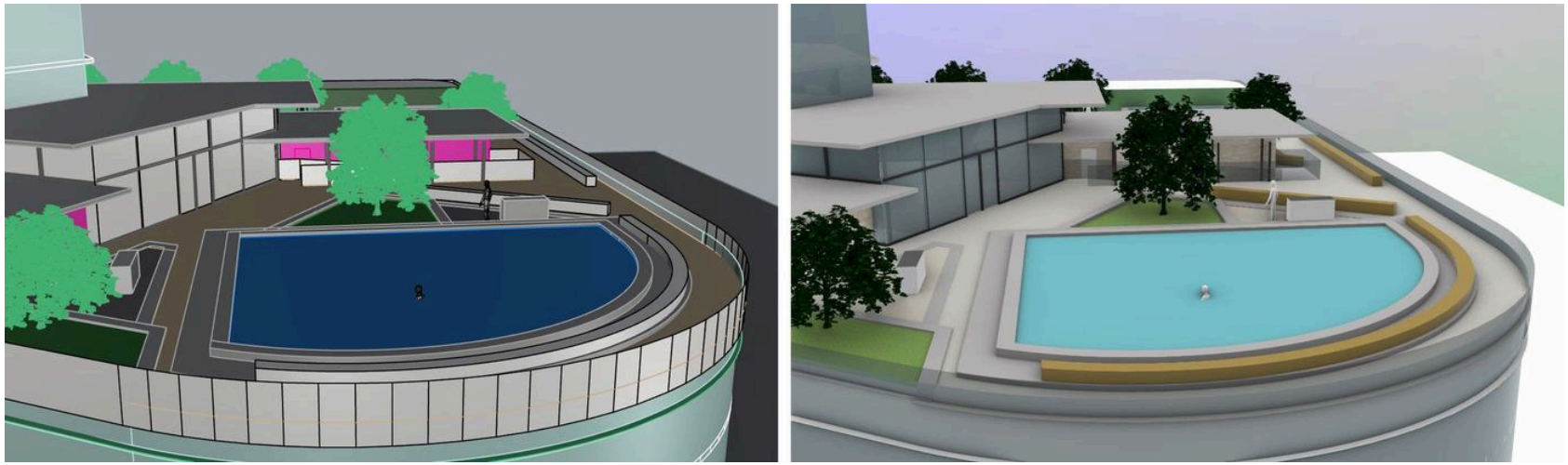
RHINO MODEL AND FINISHED RENDER | AERIAL



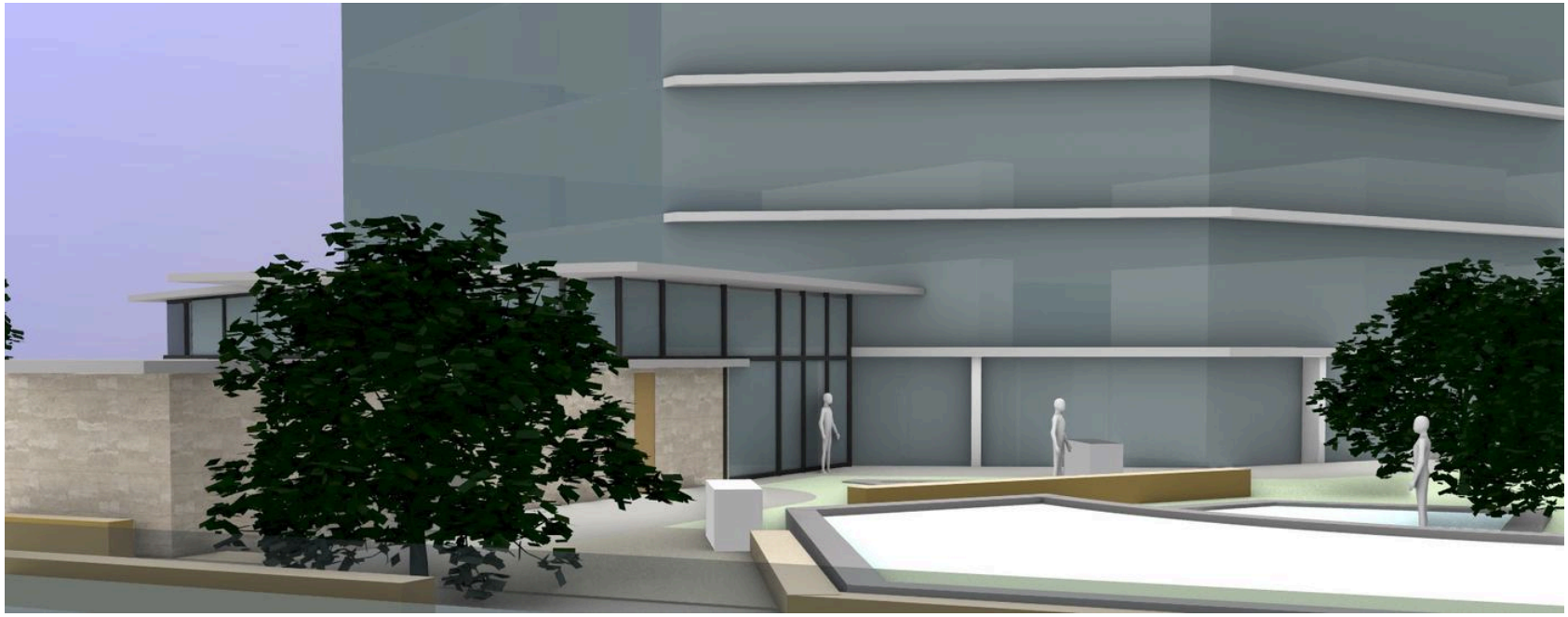


RHINO MODEL AND FINISHED RENDER | ZONE 2

THE DECK



RHINO MODEL AND FINISHED RENDER | ZONE 3



RHINO MODEL | PAVILION TERRACE

THE DECK



RENDER | PAVILION TERRACE



03

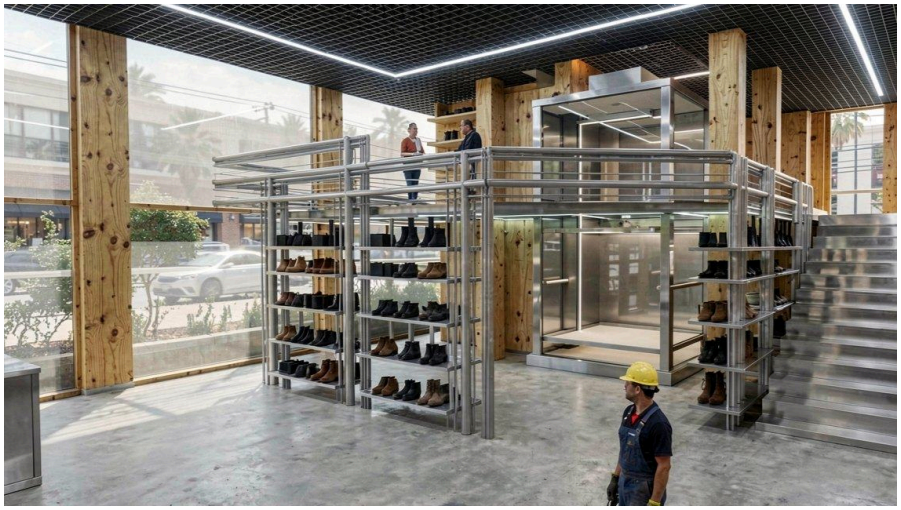
PROTOTYPE

SP26, PROTOTYPE, 3GB STUDIO

A generative pipeline for retail architecture and product design, built around a footwear brand's flagship storefront. Architectural iterations were produced through a custom Flux/ComfyUI pipeline, while the product line — a technical boot shown here in cutaway and colorway studies — was developed in parallel through a Gemini-based image pipeline. The two threads feed each other: the building's industrial material palette of raw wood, steel, and concrete is carried directly into the product's own construction and display.



CUTAWAY SECTION ELEVATION OF THE FULL BUILDING



INTERIOR RETAIL FLOOR WITH STEEL AND TIMBER SHELVING DISPLAY



INTERIOR COUNTER SCENE WITH STAFF AND CUSTOMER



TECHNICAL BOOT PRODUCT RENDER, STUDIO SHOT



BOOT COLORWAY COMPARISON SHEET, FOUR VARIANTS



TECHNICAL CUTAWAY OF THE BOOT'S INTERNAL CONSTRUCTION